

Intertemporal Consumption

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Intertemporal Consumption choice

- So far, we have considered:
 - How an individual will allocate a given amount of money over different consumption goods.
 - How an individual will allocate his time between enjoying leisure and earning money in the labor market to be used for consuming goods.
- Another thing to consider is how an individual will decide how much of his money should be consumed now, and how much he should save for consumption in the future (or how much to borrow for consumption in the present).
- To think about this, instead of considering how an individual trades off one good for another and vice versa, we can think about how an individual trades off consumption (of all goods) in the present for consumption (of all goods) in the future.

A simple model of intertemporal consumer choice

- Suppose that there is a consumer who faces a consumption choice problem across two periods.
- There are **only two periods**. But this is easily generalizable.
- In each period, there is only one good, which we denote by c_t , $t = 1, 2$.
- You can think about c_t as a composite good, i.e. dollars of consumption in all consumables.
- c_1 : Consumption in dollars, in period 1.
- c_2 : Consumption in dollars, in period 2.
- I will also refer to c_1 as **present consumption**, and c_2 as **future consumption**.
- Assume for the moment that the **market prices** of c_1 and c_2 are equal to 1.

The Budget Constraint

- Suppose that the consumer receives in each period, an **endowment of money** m_t , $t = 1, 2$.
- Thus m_1 is the present endowment of money and m_2 is the future endowment of money.
- (m_1, m_2) is exogenous. However, this assumption can be relaxed by using the model from last class.
- Examples:
 - An individual who earns \$50K each year in the labor market
 $m_t = 50K$, $t = 1, 2$
 - An individual who earns nothing but expects to inherit \$100K next year ($m_1 = 0$, $m_2 = 100K$)

The Budget Constraint

- The individual has access to financial intermediaries, for example a bank or a credit union.
- She can either lend or borrow money at an interest rate r
- Let S denote the amount this consumer saves in period 1.
- Note that if $S > 0$ the consumer is saving, if $S < 0$ the consumer is borrowing and if $S = 0$, then she is neither borrowing nor saving.
- What c_t needs to satisfy in each period? A per period budget constraint!
 - $c_1 = m_1 - S$, note that $S = m_1 - c_1$.
 - $c_2 = m_2 + (1 + r)S$
- If $S = m_1 - c_1 > 0$, the consumer is saving, and in the next period she earns on this “investment” $(1 + r)(m_1 - c_1)$, i.e. she gets the principal plus interest.
- Similarly, if $S = m_1 - c_1 < 0$, next period she needs to pay for the loan $(1 + r)(m_1 - c_1)$.

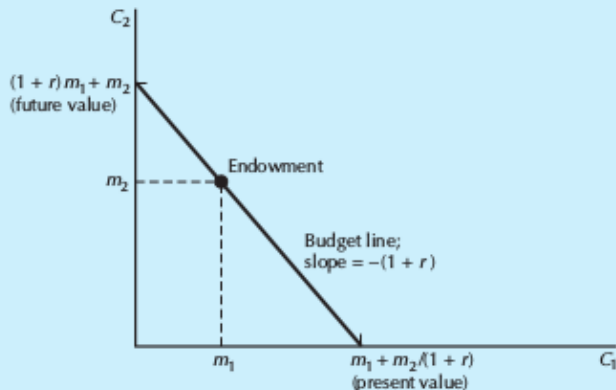
The Budget Constraint

- Recall:
 - $c_1 = m_1 - S$, note that $S = m_1 - c_1$.
 - $c_2 = m_2 + (1 + r)S$
- We can join this two budget constraint into a single consolidated budget constraint:
- $c_2 = m_2 + (1 + r)S = m_2 + (1 + r)(m_1 - c_1)$
- Thus $(1 + r)c_1 + c_2 = (1 + r)m_1 + m_2$.
- We can equivalently rewrite the budget constraint as
$$c_1 + \frac{c_2}{1+r} = m_1 + \frac{m_2}{1+r}$$
- What can we say about this budget constraint?

The Budget Constraint

- When we write the constraint in this form $(1 + r)c_1 + c_2 = (1 + r)m_1 + m_2$ is called **the future value budget constraint**.
 - Writing the constraint in this way has the interpretation that the cost of opportunity (relative price) of consumption today c_1 is $(1 + r)$, i.e. if I consume 1 unit more today, the amount of money I loose if instead I save that unit and receive interest tomorrow is $(1 + r)$ which then I can use in consuming more tomorrow.
- On the other hand, when we write the budget constraint in the form $c_1 + \frac{c_2}{1+r} = m_1 + \frac{m_2}{1+r}$ is called **the present value budget constraint**.
 - In this interpretation, the cost of opportunity (relative price) of consumption tomorrow c_2 is $1/(1 + r)$, i.e. one more unit of consumption tomorrow implies that the consumer has to save today $1/(1 + r)$ dollars instead of using it for present consumption so she can afford 1 unit tomorrow.

Graphing the Budget Constraint.



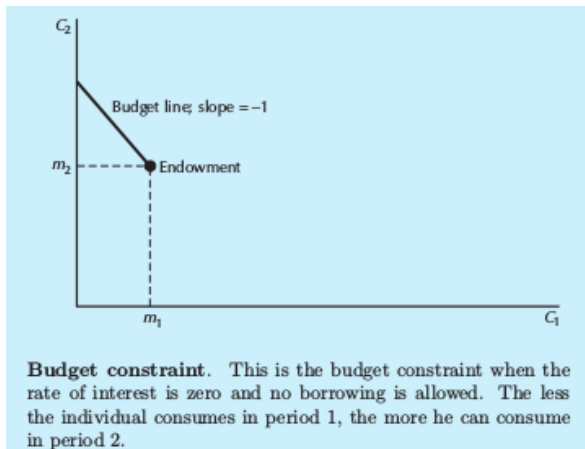
Present and future values. The vertical intercept of the budget line measures future value, and the horizontal intercept measures the present value.

Important remarks

- Notice that the point $(c_1, c_2) = (m_1, m_2)$ is always in the budget constraint. Intuitively, this means that if you neither borrow nor save, you can just use your endowments of money in each period to buy exactly that level of consumption.
- if $c_1 < m_1$ then you are a **lender**, i.e. you are saving by putting money in the bank. Is as if you were **loaning money** to the bank to get interest in return. You are at the right of the endowment point.
- if $c_1 > m_1$ then you are a **borrower**, i.e. you are asking for money from the bank in return of having to pay them tomorrow what you owe them plus interest. You are at the left of the endowment point.

A special case, $S \leq 0$ and $r = 0$

- Suppose that the consumer doesn't have access to financial intermediaries. $r = 0$.
- She can only save, but without earning any interest. $S \geq 0$.
- Like putting your money under the mattress.



Changes in r for the budget constraint.

- Now we turn to analyze what happens after changes in r .
- As pointed out before, irrespective of r , the point $(c_1, c_2) = (m_1, m_2)$ is always in the budget line.
- Furthermore, changes in r change **both intercepts** at the same time.
- Then what we have is that after r changes, the budget line rotates, and the pivot is the point (m_1, m_2)

The consumer's problem

- The consumer has preferences over streams of consumption (c_1, c_2) , i.e. preferences over present and future consumption.
- We will represent the preferences with a utility function $u(c_1, c_2)$.
- The ratio $\frac{\partial u}{\partial c_1} / \frac{\partial u}{\partial c_2}$ is now called the **intertemporal marginal rate of substitution (IMRS)**.
- We will assume that IMRS is strictly decreasing so we have strictly convex indifference curves and also that u is monotone.
- Therefore, the consumer's problem is:

$$\begin{aligned} & \max_{c_1, c_2 \in \mathbb{R}_+^2} && u(c_1, c_2) \\ & \text{subject to} && c_1 + \frac{c_2}{1+r} = m_1 + \frac{m_2}{1+r} \end{aligned} \tag{1}$$

- Given our assumptions, the optimal choices are found when

$$IMRS = (1 + r)$$

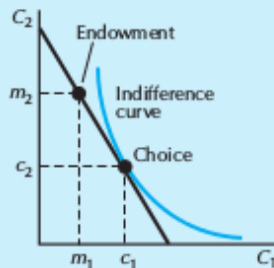
- Solving in the usual way¹, we get the per period demands (uncompensated) for each good:

$$c_1(r, m_1, m_2) \quad c_2(r, m_1, m_2)$$

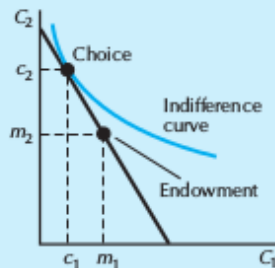
- Example $U(c_1, c_2) = c_1^\alpha c_2^{1-\alpha}$ where $\alpha \in [0, 1]$

¹Solve $IMRS = (1 + r)$ for one variable and then plug in the Budget constraint.

Comparative Statics



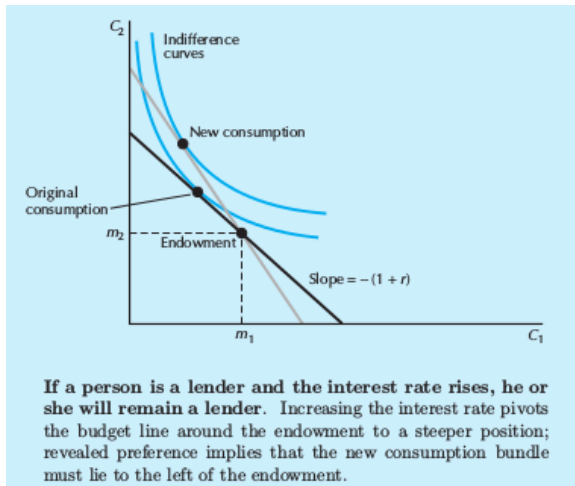
A Borrower



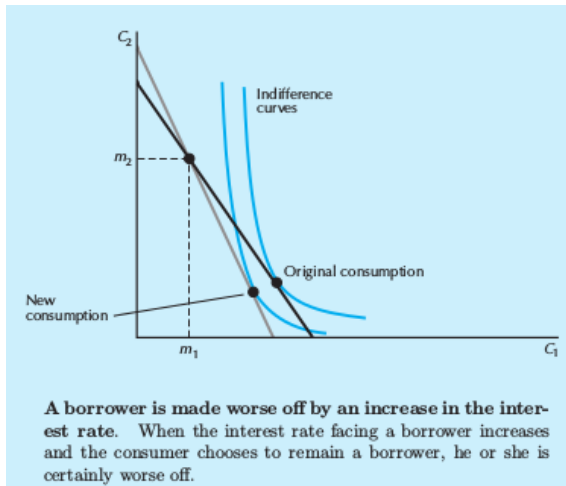
B Lender

Borrower and lender. Panel A depicts a borrower, since $c_1 > m_1$, and panel B depicts a lender, since $c_1 < m_1$.

Comparative Statics, increase in r



Comparative Statics, increase in r



Summary

- For an increase in r
 - A consumer that was a lender remains a lender and is made better off.
 - A consumer that was a borrower remains a borrower and is made worse off.
- For a decrease in r
 - Ambiguous results, a lender might decide to become a borrower and a borrower might decide to become a lender.

The Slutsky equation

- Notice that now the level of income of the consumer I actually depends on the “price”.
- In this case, total income, using the future value budget constraint, is $I = (1 + r)m_1 + m_2$
- Notice that $\partial I / \partial r = m_1$
- Thus we have:

$$\frac{\partial c_1(r, I)}{\partial r} = \frac{\partial c_1(r, m_1, m_2)}{\partial r} + \frac{\partial c_1(r, m_1, m_2)}{\partial I} \frac{\partial I}{\partial r}$$

- But this is equal to

$$\frac{\partial c_1(r, I)}{\partial r} = \frac{\partial c_1(r, m_1, m_2)}{\partial r} + \frac{\partial c_1(r, m_1, m_2)}{\partial I} m_1$$

- Using now the Slutsky equation on the term $\frac{\partial c_1(r, m_1, m_2)}{\partial r}$ yields

$$\frac{\partial c_1(r, I)}{\partial r} = \frac{\partial c_1^C}{\partial r} + (m_1 - c_1) \frac{\partial c_1}{\partial I}$$